

115.109

Introduction to Business Analytics

Topic 4: Variation



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(Please contact me via Keeping in
Touch in Stream)



Measures of Variation / Dispersion

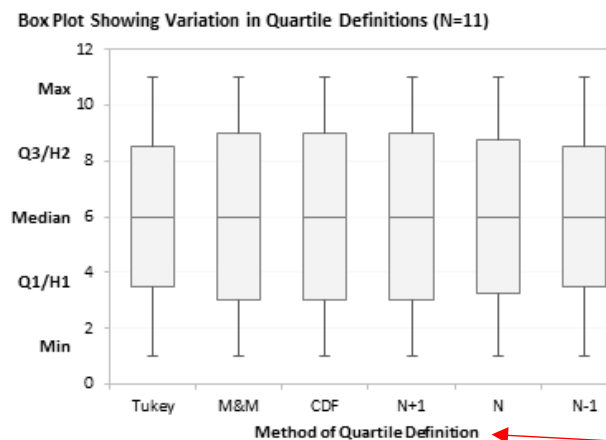
- Range or spread
- Minimum and Maximum
 - range = max – min
- Quartiles
 - Lower Quartile: 25%ile (bottom)
 - Upper Quartile: 75%ile
 - InterQuartile Range: UQ – LQ
- Deciles (10%iles)
- Deviation

'N+1' method
used by Excel:
=Quartile.Exc(range,1)

$$LQ = \frac{n+1}{4}$$

$$UQ = \frac{3 \times (n+1)}{4}$$

'N+1' method
used by Excel:
=Quartile.Exc(range,3)



NB: the “.Exc” means the median is EXCLUDED when calculating the LQ and UQ.

The alternative =Quartile.Inc() INCLUDES the median in each half (it is represented twice). Different stats packages use either method, but the “.Inc” version is more common. There are other methods too!

*M&M: Moore & McCabe; CDF: Cumulative Distribution Function



Ranges

6 + 7 + 11 + 16 + 17 + 20 + 25
+ 30 +

$N = 15$

31 + 32 + 33 + 34 + 34 + 37 + 41

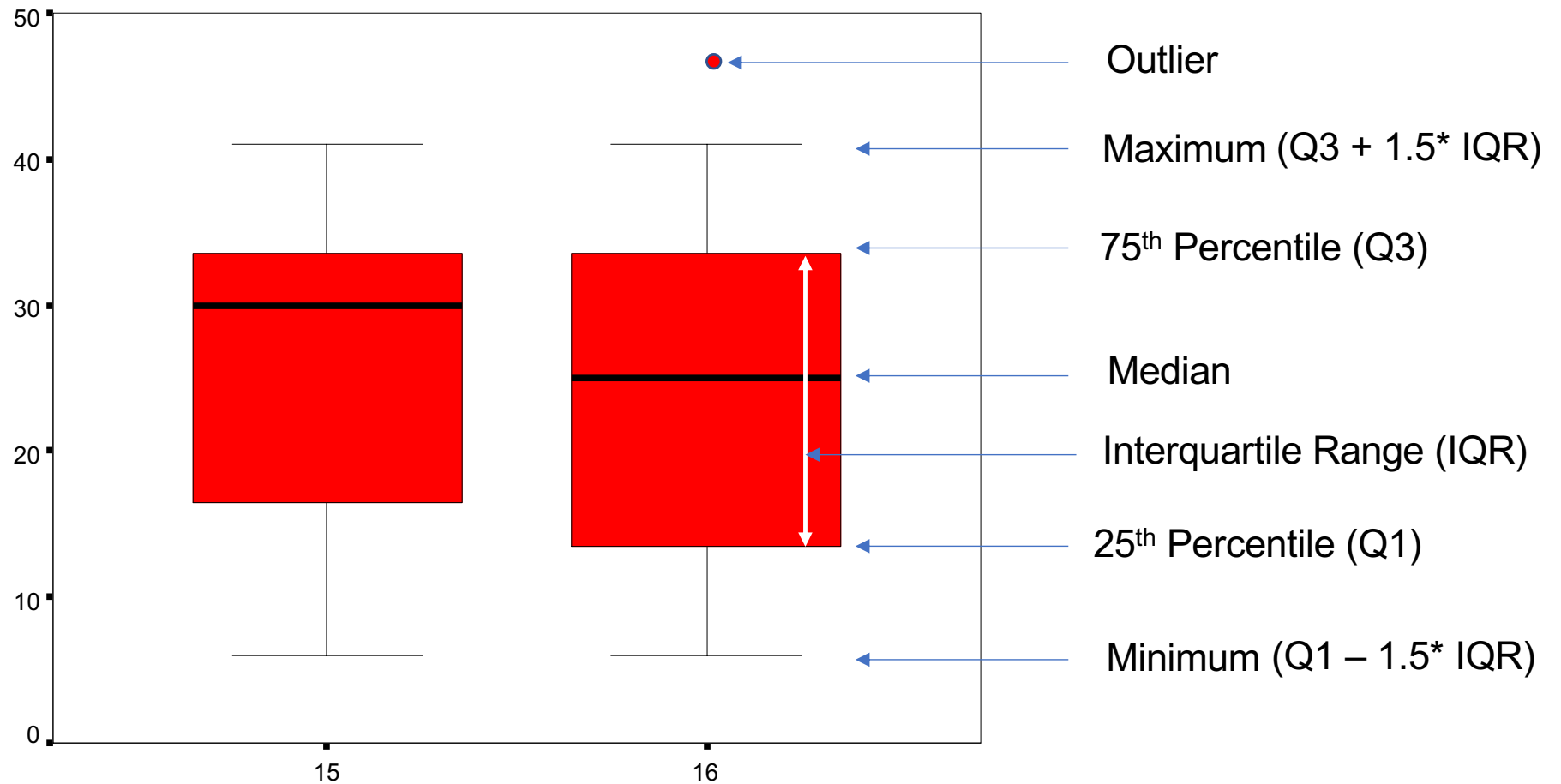
- Range: $= 41 - 6 = 35$
- Median: 30
- InterQuartile Range: $= 34 - 16 = 18$

$$LQ = \frac{15+1}{4} = 4th$$

$$UQ = \frac{3 \times (15+1)}{4} = 12th$$



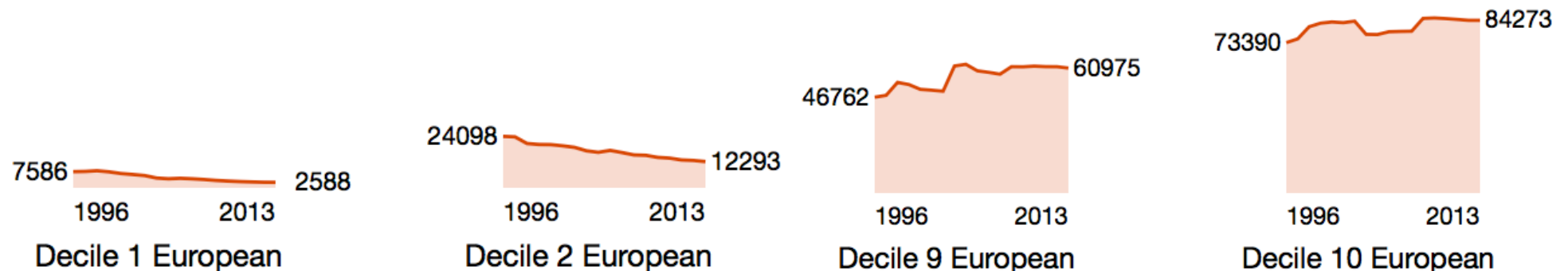
Dispersion: Boxplot





School Deciles

- Deciles of schools, not the student population
- “The Ministry of Education uses a decile rating (ranking) system for school funding purposes. Each decile contains approximately 10% of schools. Schools in decile 1 have the highest proportion of students from low socio-economic backgrounds. Schools in decile 10 have the lowest proportion of these students.” (<http://www.ero.govt.nz/Help/FAQs>)





Customer Deciles: Perceptive



Spend vs growth

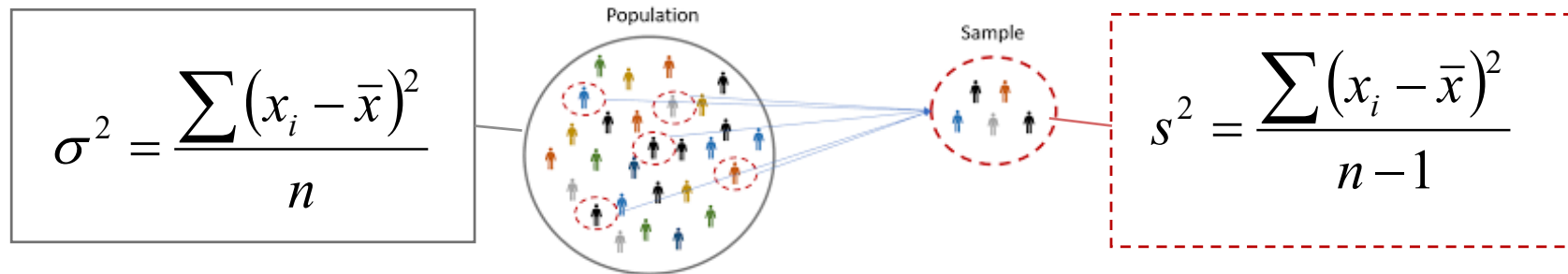
Y1 Spend Percentile	Y1 Spend Upper Limit	Avg Y1 Spend	Avg Y2 Spend	Avg Growth	Avg Growth %
10%	\$16.50	\$11.59	\$90.43	\$78.85	681%
20%	\$24.99	\$20.86	\$84.79	\$63.92	306%
30%	\$34.99	\$30.05	\$94.39	\$64.33	214%
40%	\$47.98	\$41.27	\$108.14	\$66.87	162%
50%	\$65.00	\$56.52	\$112.89	\$56.37	100%
60%	\$87.99	\$75.64	\$132.80	\$57.16	76%
70%	\$123.97	\$105.09	\$146.91	\$41.82	40%
80%	\$180.99	\$150.77	\$189.54	\$38.77	26%
90%	\$305.48	\$234.44	\$256.83	\$22.40	10%
100%	\$5,000.00	\$634.94	\$602.86	(\$32.09)	-5%
Total		\$184.06	\$221.40	\$37.34	20%

Based on spend by customer in year 1: see <https://webcast.massey.ac.nz/Mediasite/Play/bdb70603063f4da3bb732235743d11e41d>



Deviation

- Variance
 - Measure of variation in data around the mean
 - σ^2 for populations, s^2 for samples



- Standard Deviation
 - Measure of variation in data around the mean
 - σ for populations, s for samples

$$s = \sqrt{s^2} \qquad s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

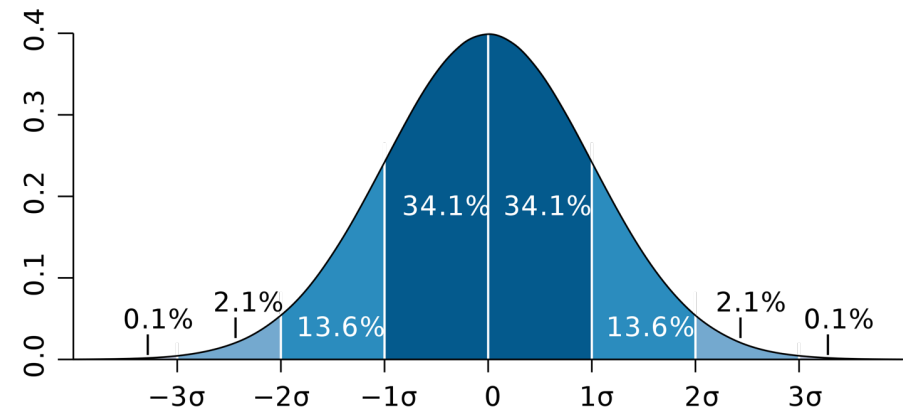
Increases Std. Dev



Deviation

- Standard deviation: how much the observations in the sample vary from the mean
- Standard error of the mean: how much sample means from the same population vary
- *Estimate* of the standard error of the mean:

$$s.e.(\bar{x}) = \frac{s}{\sqrt{n}}$$





Standard Deviation

- The most commonly used measure of variation is the *standard deviation*
- It takes into account every observation and measures the ‘average deviation’ of observations from the mean
- It works with squares of residuals, not absolute values (like mean deviation), therefore it is easier to use in further calculations
- The values of the mean deviation and standard deviation should be reasonably close, since they are both measuring the variation of the observations from their mean



Standard Deviation: Population

- *Population standard deviation*
 - *Uses squares* of the residuals, which will eliminate the effect of the signs, since squares of numbers cannot be negative
- Step 1: Find the sum of the squares of the residuals
- Step 2: Find their mean
- Step 3: Take the *square root* of this mean

$$\text{Standard deviation} = \sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{N}}$$

Where N = the size of the population

The *square* of the population standard deviation σ is called the *population variance*

$$\text{Variance} = \sigma^2$$



Standard Deviation: Sample

- *Sample standard deviation*
 - It is rare to calculate the value of σ since populations are usually very large and difficult to measure
 - It is far more likely that the *sample standard deviation* (denoted by s) will be needed

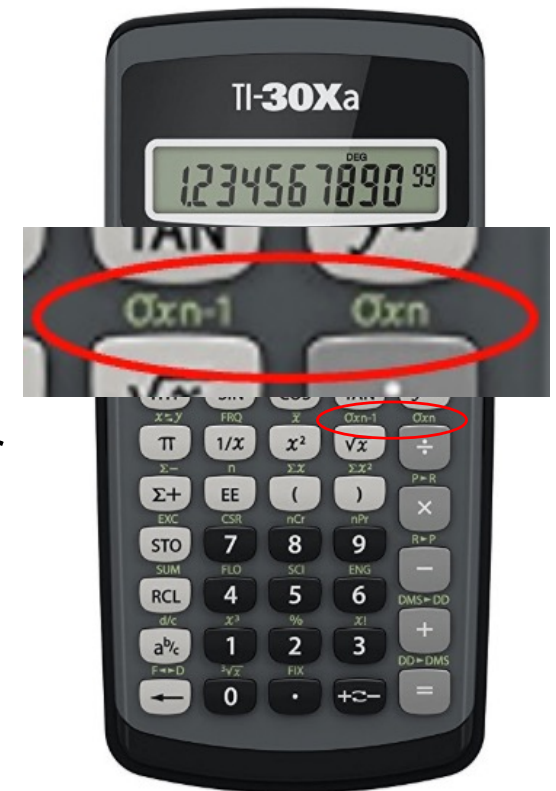
$$\text{Sample standard deviation} = s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

- Where: $(n - 1)$ is the number of observations in the sample



Standard Deviation

- *A note on the use of $(n - 1)$ in formulae*
 - If the value of n is large, it will only make a slight difference to the answer whether you divide by n or $(n - 1)$
 - To calculate the value of s from a *sample* the calculator button will usually be indicated by one of σ_{n-1} or $x\sigma_{n-1}$ or s_x or s written either on it or near it
 - To calculate the value of s from a *population*, the calculator key will usually be indicated by one of σ_n or $x\sigma_n$ or σ_x or σ written either on it or near it





Standard Deviation: Key points

- Standard deviation is in the same units as the mean
 - E.g if the it is mean age (in years), the st dev is also in years;
 - if mean spend in store, the st dev is also in \$
- The *square* of the standard deviation is called the *variance* (which is not in the same units as the mean)
- The standard deviation cannot be negative
- The standard deviation of a set of data is zero if, and only if, the observations are of equal value
- The standard deviation can *never* exceed the range of the data
- The more scattered the data, the greater the standard deviation



Standard Deviation: calculation

Step 1: Calculate the mean $\bar{x}$

Step 2: For each x-value, find the value of the residual $x - \bar{x}$

Step 3: Square the residuals $(x - \bar{x})^2$

Step 4: Calculate the sum of the squares of the residuals $\sum (x - \bar{x})^2$

Step 5: Divide the sum found in Step 4 by $(n - 1)$

Step 6: Take the square root of the quantity found in Step 5:

this is the sample standard deviation

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$



Deviation

$6 + 7 + 11 + 16 + 17 + 20 + 25$
 $+ 30 +$
 $31 + 32 + 33 + 34 + 34 + 37 + 41$
 $N = 15 \quad \bar{x} = 24.9$

Variance
 $= 127.6$

Standard deviation:
 $= \text{square root } (127.6)$
 $= 11.3$

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

	x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
	6	-18.9	358.5
	7	-17.9	321.6
	11	-13.9	194.1
	16	-8.9	79.8
	17	-7.9	62.9
	20	-4.9	24.3
	25	0.1	0.0
	30	5.1	25.7
	31	6.1	36.8
	32	7.1	49.9
	33	8.1	65.1
	34	9.1	82.2
	34	9.1	82.2
	37	12.1	145.6
	41	16.1	258.1
N	15		
Σ	374	0.0	1786.9
$\bar{x}$	24.93		
$\Sigma(x_i - \bar{x})^2 / (n-1)$ aka Variance			127.6
sqrt			11.2977031
stdev()			11.2977031



St. Dev. change after adding an additional observation

$$\begin{aligned} &6 + 6 + 7 + 11 + 16 + 17 + 20 \\ &\quad + 25 + 30 + \\ &31 + 32 + 33 + 34 + 34 + 37 + 41 \end{aligned}$$

$$N = 16 \quad \bar{x} = 23.75$$

$$\bar{x} : \cancel{24.93} \rightarrow 23.75$$

Variance

$$= \cancel{127.6} \rightarrow 141.5$$

Standard deviation:

$$= \text{square root } (141.5)$$

$$= \cancel{11.3} \rightarrow 11.9$$

	x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
	6	-17.8	315.1
	6	-17.8	315.1
	7	-16.8	280.6
	11	-12.8	162.6
	16	-7.8	60.1
	17	-6.8	45.6
	20	-3.8	14.1
	25	1.3	1.6
	30	6.3	39.1
	31	7.3	52.6
	32	8.3	68.1
	33	9.3	85.6
	34	10.3	105.1
	34	10.3	105.1
	37	13.3	175.6
	41	17.3	297.6
N	16		
Σ	380	0.0	2123.0
$\bar{x}$	23.75		
$\Sigma(x_i - \bar{x})^2 / (n-1)$ aka Variance			141.5
sqrt			11.8967783
stdev()			11.8967783



Strongly disagree 1 2 3 4 5 6 7 Strongly agree

☒ ☐ ☐ ☐ ☐ ☐ ☐

[illegible]



Deriving $\bar{X}$ and s manually: Calculating the Mean

X		f		$f x$
1	*	0	=	0
2	*	0	=	0
3	*	4	=	12
4	*	15	=	60
5	*	34	=	170
6	*	43	=	258
7	*	52	=	364
N / Sum		148		864
$\bar{X} / s$				<u>5.83783784</u>



Deriving $\bar{X}$ and s manually: Calculating Residuals

X	f	fx	$(X - \bar{X})$
1	0	0	-4.83783784
2	0	0	-3.83783784
3	4	12	-2.83783784
4	15	60	-1.83783784
5	34	170	-0.83783784
6	43	258	0.16216216
7	52	364	1.16216216
N / Sum	148	864	
$\bar{X} / s$		5.83783784	



Deriving $\bar{X}$ and s manually: Calculating Squared Residuals

X	f	fx	$(X-\bar{X})$	$(X-\bar{X})^2$
1	0	0	-4.83783784	23.40467495
2	0	0	-3.83783784	14.72899927
3	4	12	-2.83783784	8.053323594
4	15	60	-1.83783784	3.377647918
5	34	170	-0.83783784	0.701972243
6	43	258	0.16216216	0.026296567
7	52	364	1.16216216	1.350620891
N / Sum	148	864		
$\bar{X} / s$		5.83783784		



Deriving $\bar{X}$ and s manually: Sum the Squared Residuals

X	f	f x	(X- $\bar{X}$)	(X- $\bar{X}$) ²	f (X- $\bar{X}$) ²
1	0	0	-4.83783784	23.40467495	0
2	0	0	-3.83783784	14.72899927	0
3	4	12	-2.83783784	8.053323594	32.21329438
4	15	60	-1.83783784	3.377647918	50.66471877
5	34	170	-0.83783784	0.701972243	23.86705625
6	43	258	0.16216216	0.026296567	1.130752374
7	52	364	1.16216216	1.350620891	70.23228634
N / Sum	(148 -1)	864			178.1081081
$\bar{X}$ / s		5.83783784		Variance	<u>1.211619782</u>
					↓ √
				Std. Dev.	<u>1.100736018</u>



Deriving $\bar{X}$ and s manually: Formulas

X	f	f x	(X- $\bar{X}$)	(X- $\bar{X}$) ²	f (X- $\bar{X}$) ²
1	0	0	-4.83783784	23.40467495	0
2	0	0	-3.83783784	14.72899927	0
3	4	12	-2.83783784	8.053323594	32.21329438
4	15	60	-1.83783784	3.377647918	50.66471877
5	34	170	-0.83783784	0.701972243	23.86705625
6	43	258	0.16216216	0.026296567	1.130752374
7	52	364	1.16216216	1.350620891	70.23228634
N / Sum	148	864		178.1081081	
$\bar{X}$ / s		5.83783784		1.100736019	

Q10 Gender	Imp_Quality
1=Male	1=Not
2=Female	important
	4=Neutral
	7=Very
	important
2	7
2	5
1	6
2	6
$\bar{X}$	5.83783784
s	1.100736019

= average()
= stdev()

$$\begin{aligned}\bar{X} &= \frac{\sum f x}{\sum f} \\ &= \frac{864}{148} \\ &= 5.838\end{aligned}$$

$$\begin{aligned}s &= \sqrt{\frac{\sum f (x - \bar{x})^2}{\sum f - 1}} \\ &= \sqrt{\frac{178.108}{148 - 1}} = 1.101\end{aligned}$$



Standard Deviation: Categorical Data

...and where each category covers a range of observations

- When calculating s from a grouped frequency distribution, we should assume that the observations in each class interval are concentrated at the *midpoint* of the interval

$$s \approx \sqrt{\frac{\sum f(m - \bar{x})^2}{\sum f - 1}}$$

$$\bar{x} = \frac{\sum fm}{\sum f}$$

where: $\bar{x}$ = the estimated mean of the grouped data
 m = the midpoint of the class interval
 f = the frequency of the class interval

$\sum fm$ = the sum of the midpoint of a class interval and that class interval's frequency

$\sum f$ = the sum of the frequencies

For example:

Age Groups 2020	
15 -19	
20 - 29	
30 - 39	
40 - 49	
50 - 59	
60 - 69	
70 - 79	
80 - 84	
85 +	

Grouped data – Calculating Mean from Class Midpoints

Data in Ordered Array:

12, 13, 17, 21, 24, 24, 26, 27, 27, 30, 32, 35, 37, 38, 41, 43, 44, 46, 53, 58

Class	Midpoint (m)	Frequency (f)	<i>fm</i>	
10 – 20	15	3 = 15 x 3 →	45	$\bar{x} = \frac{\sum fm}{\sum f} = \frac{660}{20} = 33$
20 – 30	25	6	150	
30 – 40	35	5	175	
40 – 50	45	4	180	
50 – 60	55	2	110	
Total		Σf 20	Σfm 660	

Grouped data – Calculating Mean & StDev from Midpoints

Data in Ordered Array:

12, 13, 17, 21, 24, 24, 26, 27, 27, 30, 32, 35, 37, 38, 41, 43, 44, 46, 53, 58

Class	Midpoint (m)	Frequency (f)	Mean ($\bar{x}$)	residuals $f(m-\bar{x})^2$
10 – 20	15	3	33	$3(15-33)^2$ 972
20 – 30	25	6	33	$6(25-33)^2$ 384
30 – 40	35	5	33	$5(35-33)^2$ 20
40 – 50	45	4	33	$4(45-33)^2$ 576
50 – 60	55	2	33	$2(55-33)^2$ 968
Total		Σf 20	Σfm 660	$\Sigma f(m-\bar{x})^2$ 2920

$$\bar{x} = \frac{\sum fm}{\sum f} = \frac{660}{20} = 33$$

$$s \approx \sqrt{\frac{\sum f(m-\bar{x})^2}{\sum f - 1}}$$

$$s = \sqrt{\frac{2920}{19}} = 12.4$$



Coefficient of Variation

- A measure of relative variability used to:
 - measure changes that have occurred in a population over time
 - compare variability of two populations that are expressed in different units of measurement
- It is expressed as a percentage rather than in the units of the particular data

$$V = 100 \left(\frac{s}{\bar{x}} \right) \%$$

where: $\bar{x}$ = the mean of the sample
s = the standard deviation of the sample

Calculate the coefficient of variation for the price of 400 g cans of pet food, given that the mean is 81 cents and s = 6.77 cents. Interpret the results.

$$\begin{aligned} V &= 100 \left(\frac{s}{\bar{x}} \right) \% \\ &= 100 \left(\frac{6.77}{81} \right) \% \\ &= 8.36\% \end{aligned}$$

This means that the std. dev. of the price of a 400g can is 8.36% of the mean price



Variation Summary

- The standard deviation is the most frequently used measure of variation, and because of its mathematical properties it has widespread use in problems involving statistical inference
- If the mean cannot be calculated, neither can the standard deviation
- Its value is affected by the value of every observation in the data
- If the data have a number of extreme values, the value of the standard deviation may be distorted so as not to be a good 'representative' measure of variation

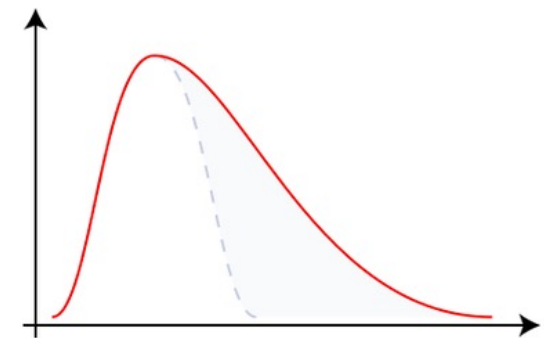


Skew

- The *skewness* of a distribution is measured by the symmetry of the ‘tails’:
a ‘long tail’ indicates skew*

(* there are many variations: a ‘short fat’ tail on one side can balance a ‘long, thin’ tail on the other)

- Distribution is *symmetrical*
 $Mean \simeq Median \simeq Mode$
- Distribution *positively skewed* (“right tailed”)
 - $Mean > Median$ (tendency not rule)
 - $Mode < Median < Mean$
- Distribution *negatively skewed* (“left tailed”)
 - $Mean < Median$
 - $Mode > Median > Mean$



Positive Skew



Negative Skew



Skew

- The *skewness* of a distribution is measured by the symmetry of the ‘tails’:
a ‘long tail’ indicates skew*

(* there are many variations: a ‘short fat’ tail on one side can balance

- Distribution is *symmetrical*
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